

COE 1195: Advanced Digital Design

Lab 4 – 32-bits MIPS CPU

Milestone 2

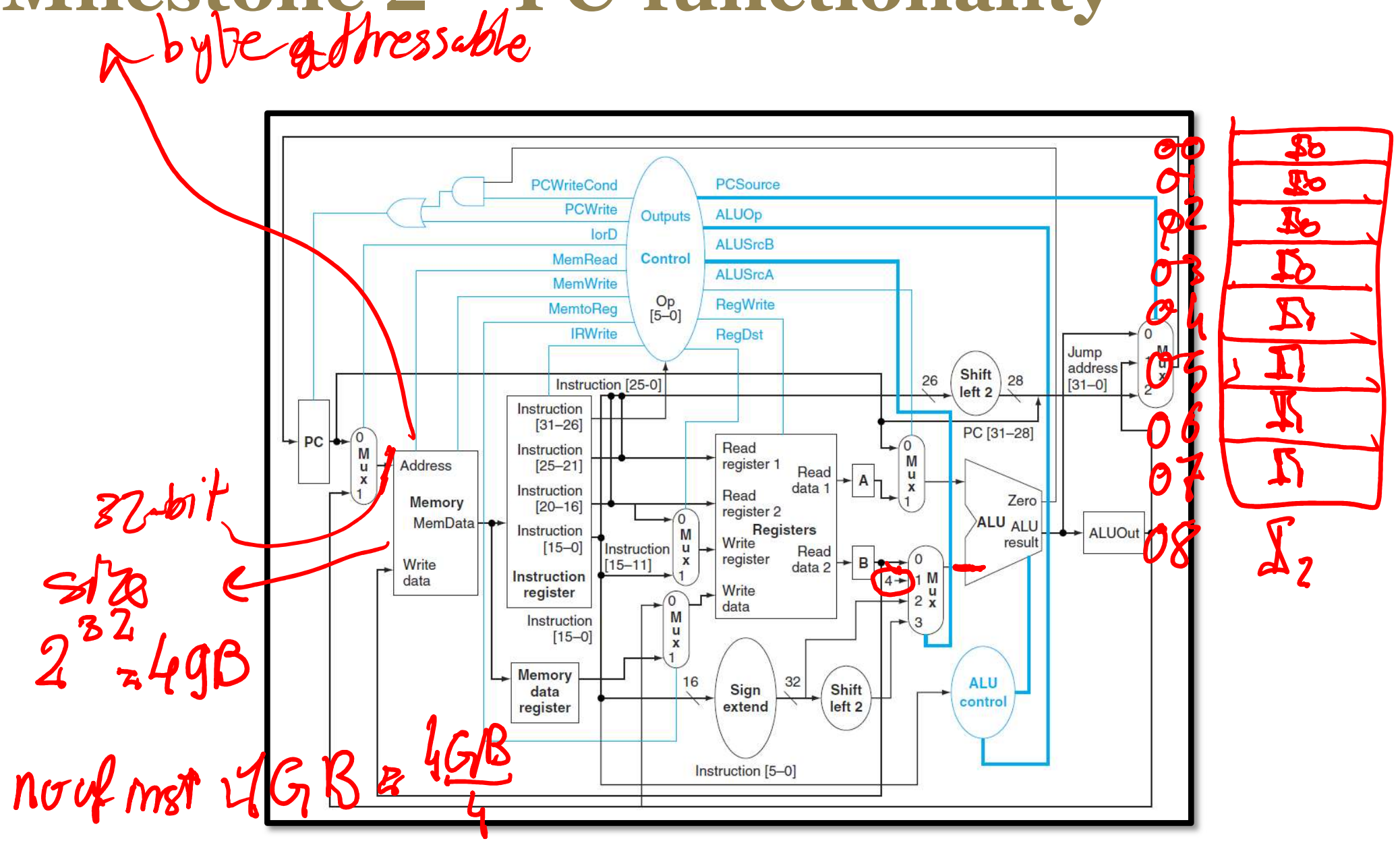
Dr. Amr Mahmoud

Milestone 2

1. Implement PC functionality

2. Implement I-Type instructions

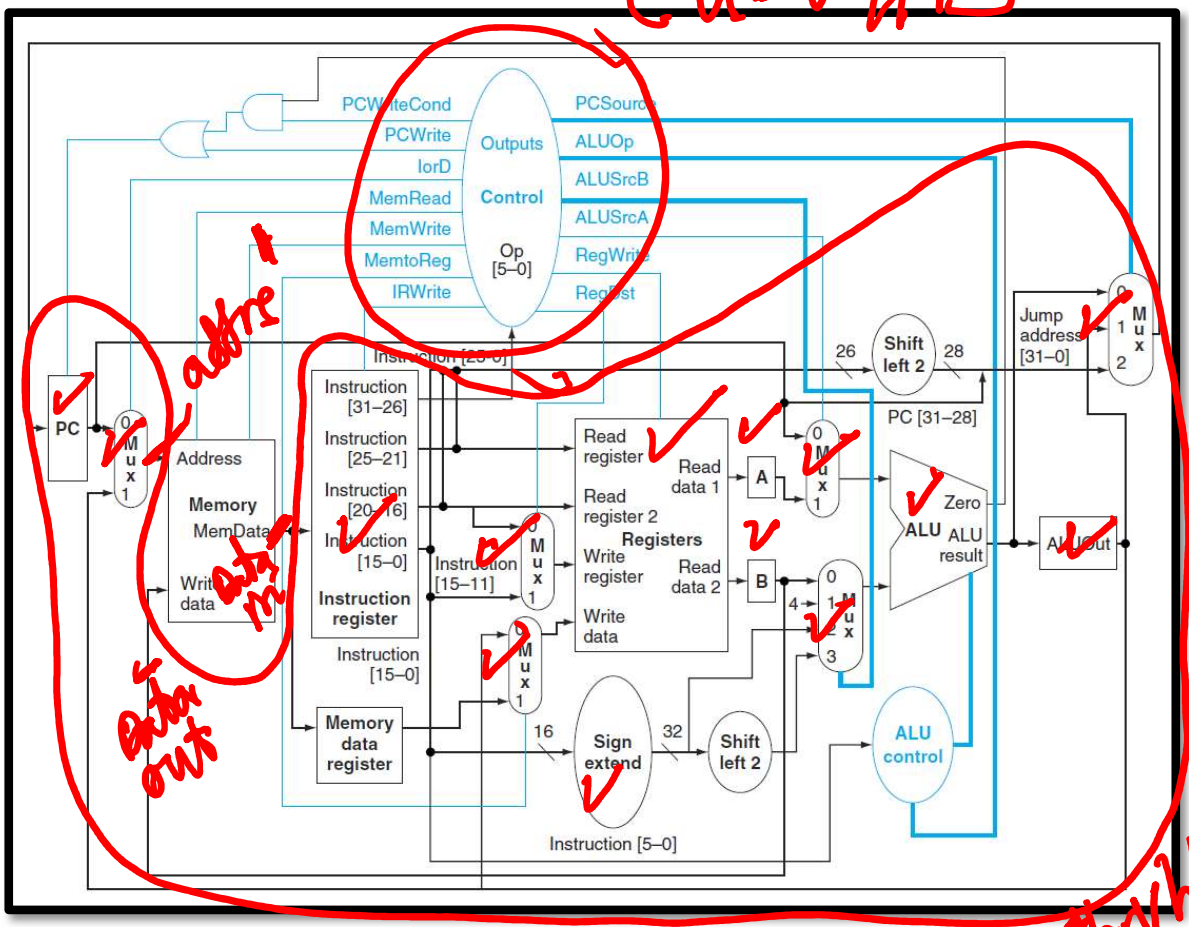
Milestone 2 – PC-functionality



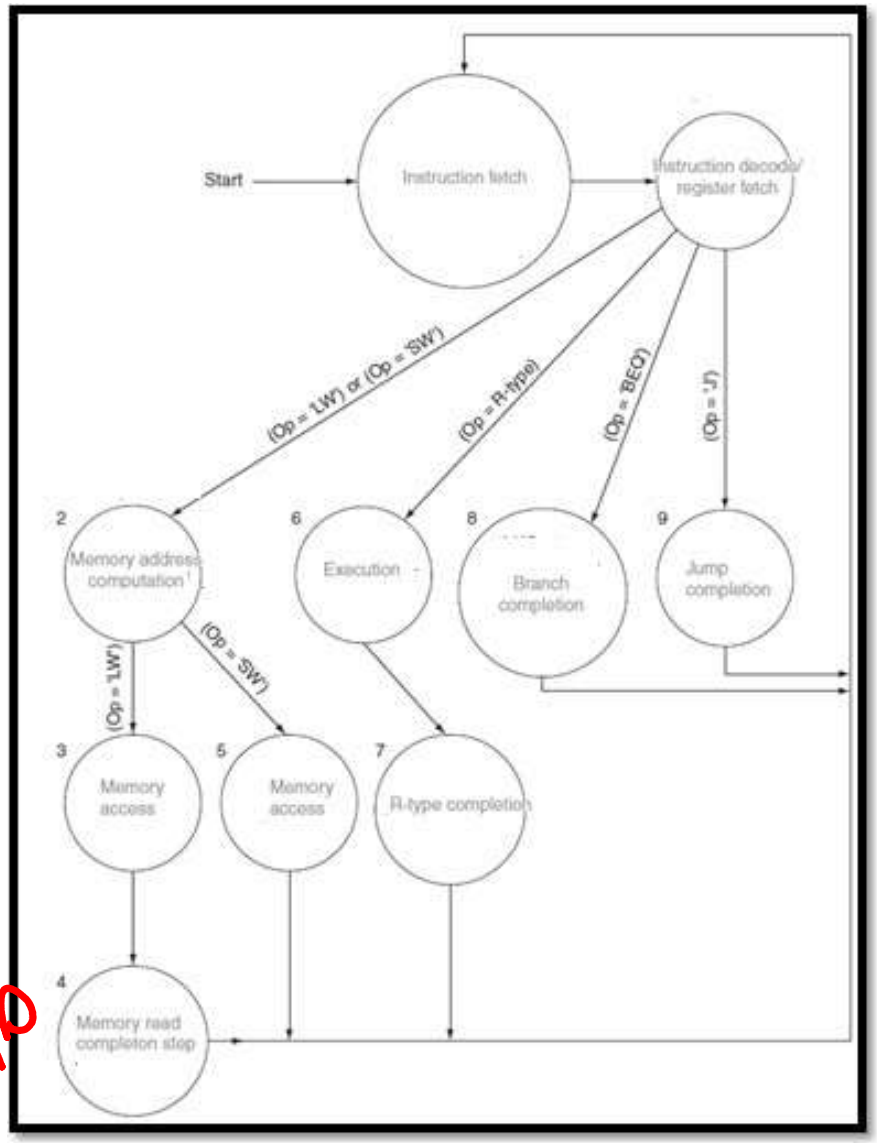
MIPS Multicycle CPU Datapath

Datapath and Control

MIPS Multicycle CPU FSM



MIPS Multicycle CPU Datapath

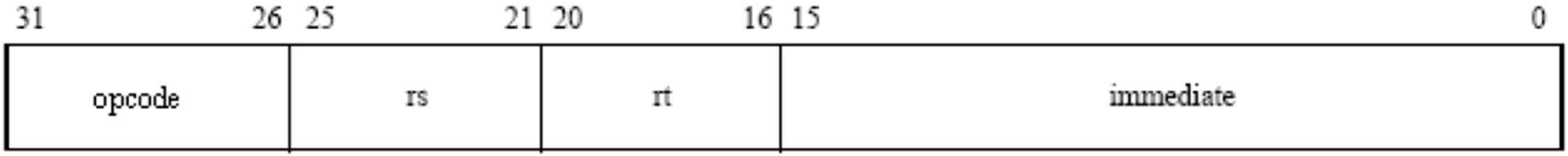


MIPS Multicycle CPU FSM Controller

I-Type Instructions

X = Y + 3

I-Type Instruction Format



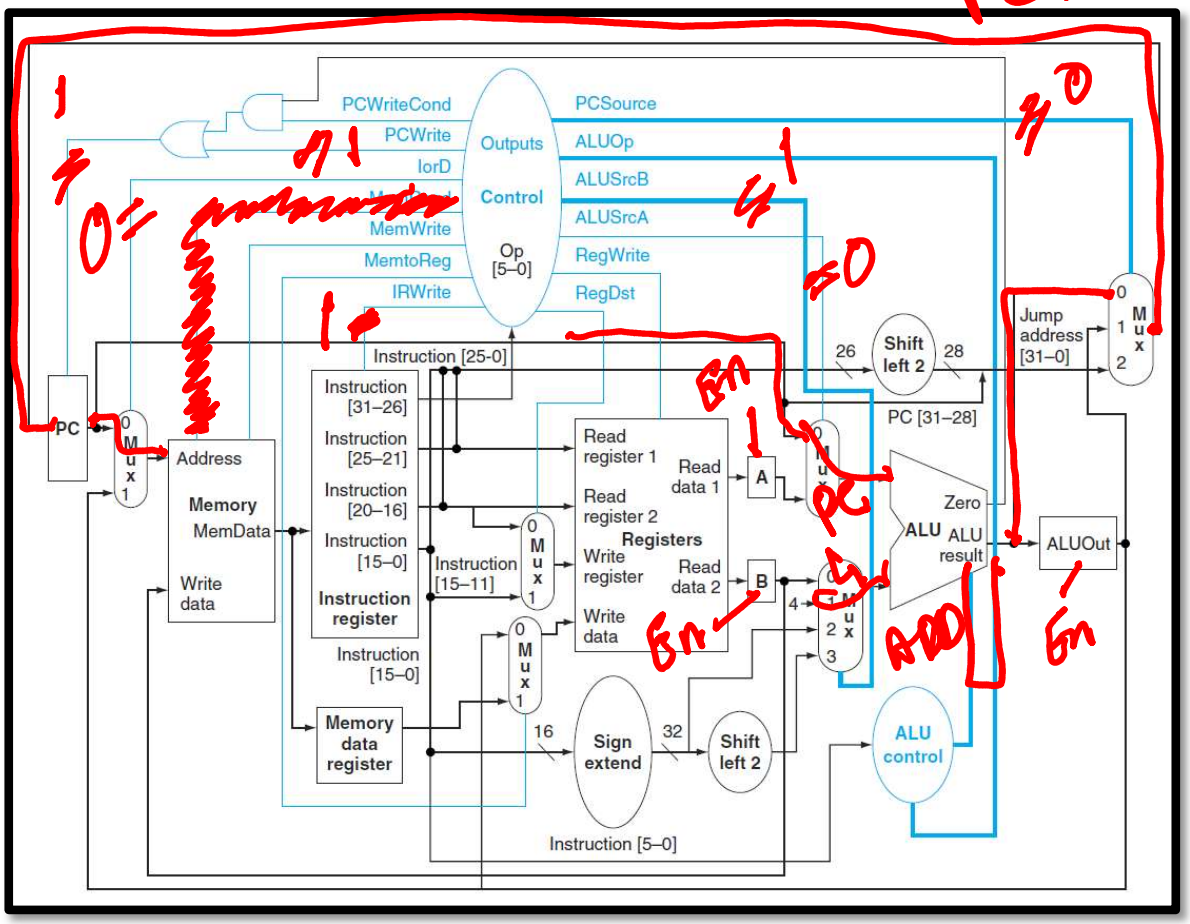
ADDI r_t, r_s, Immediate

I-Type Instructions

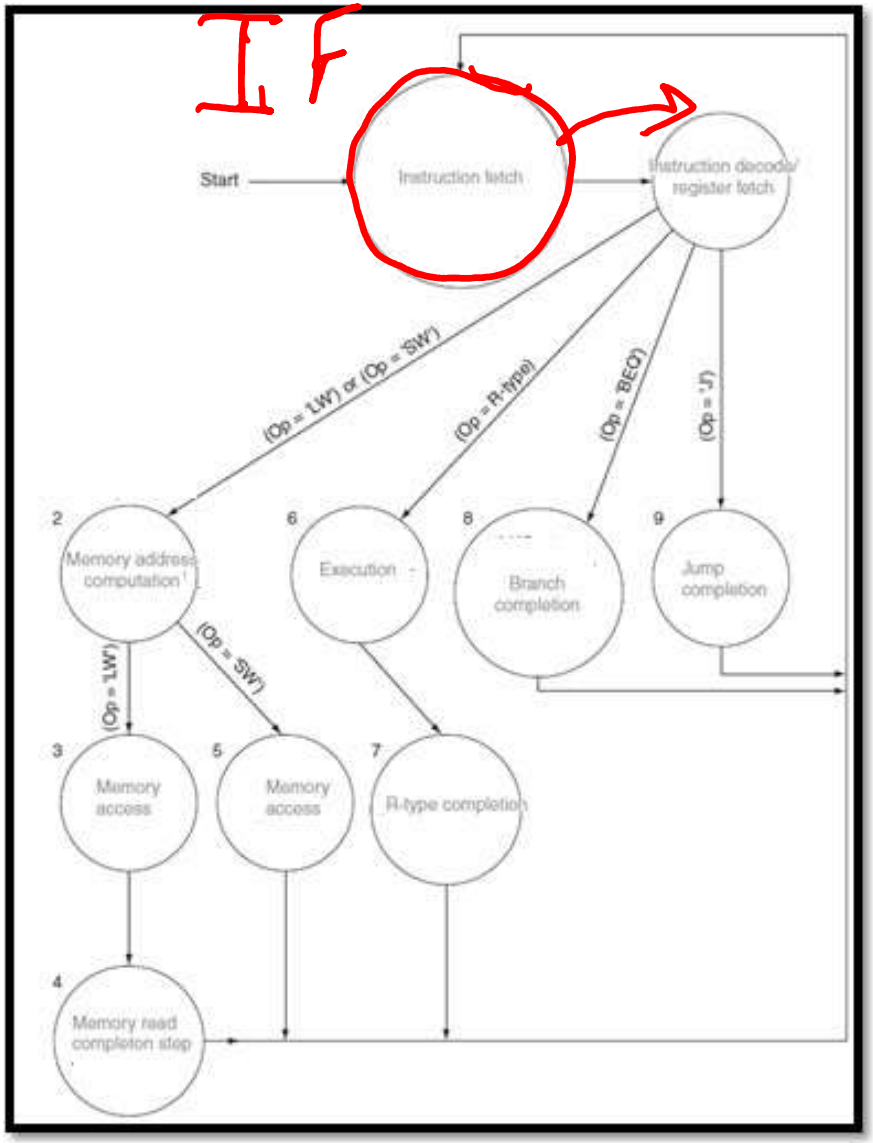
ADDI	ADDU	AND
ORI	SUB	SRA
SLLV	SLL	SLTI
LUI	SW	LW
LH	LB	CLO
BNE	BLTZAL	J
JR	MULTU	MFHI
MFLO		

Datapath and Control

MIPS Multicycle CPU FSM



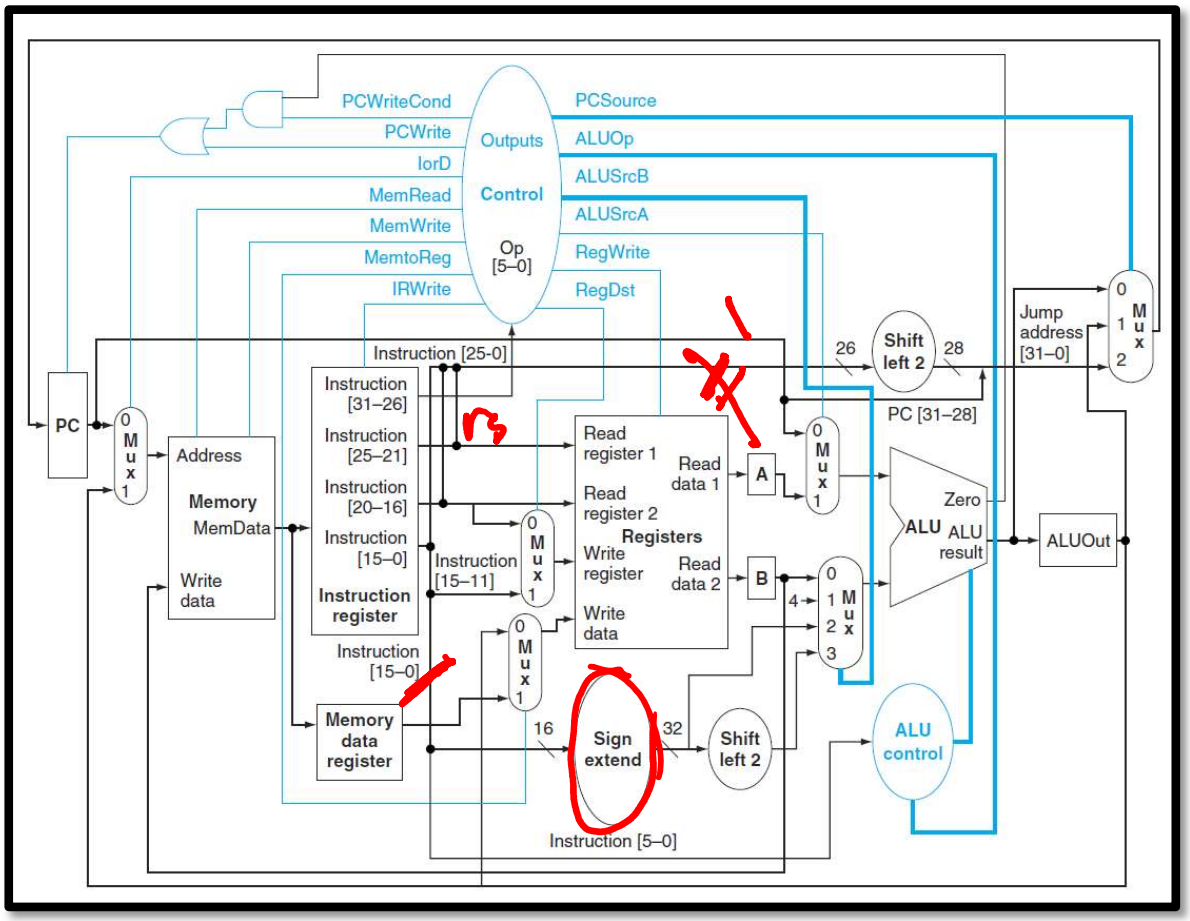
MIPS Multicycle CPU Datapath



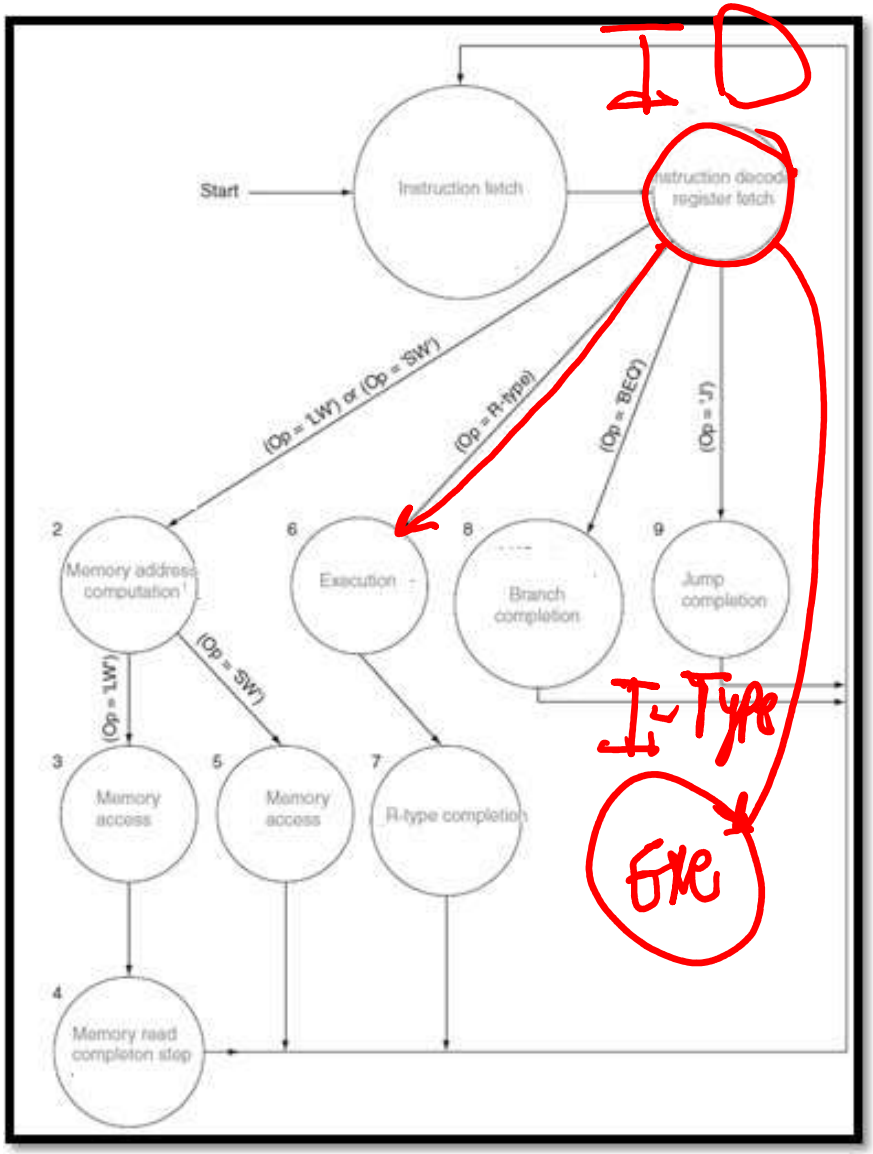
MIPS Multicycle CPU FSM Controller

Datapath and Control

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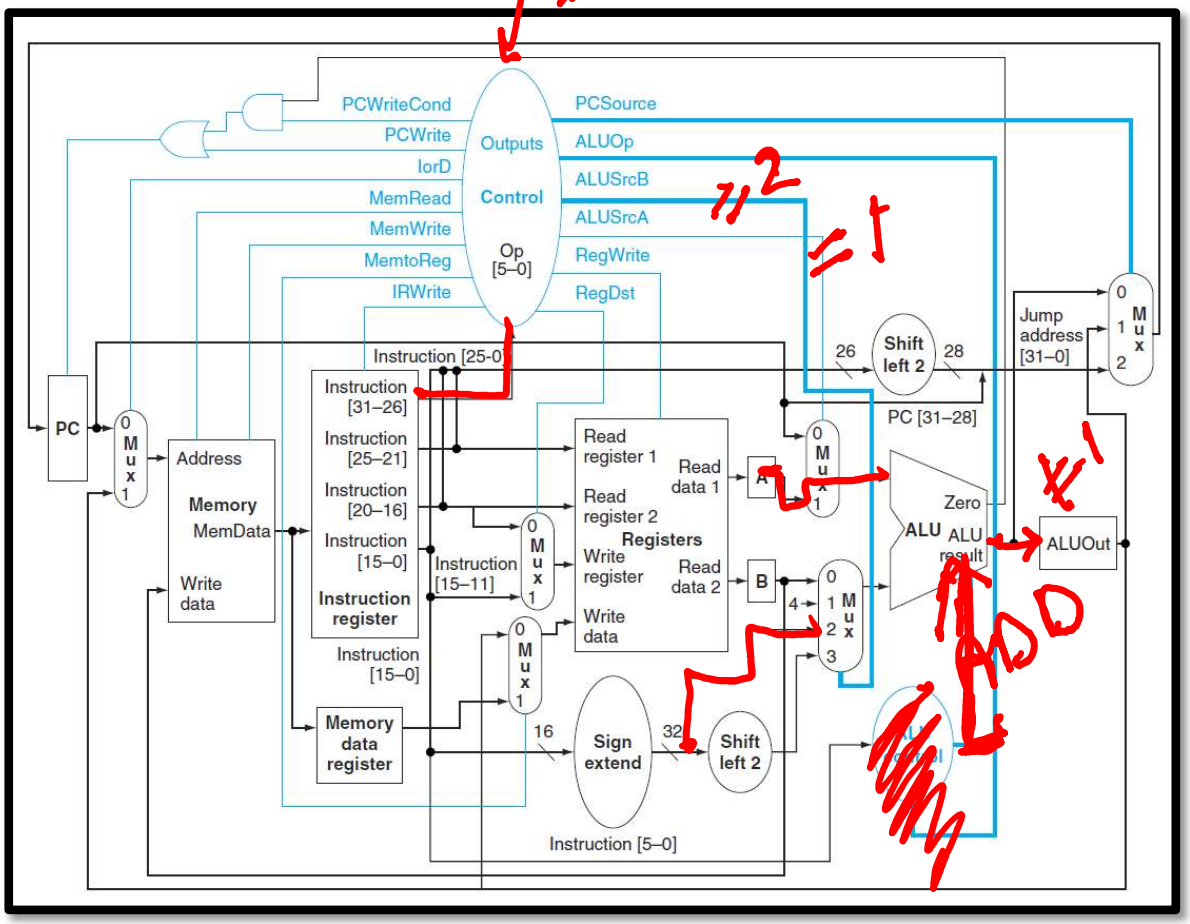
MIPS Multicycle CPU Datapath



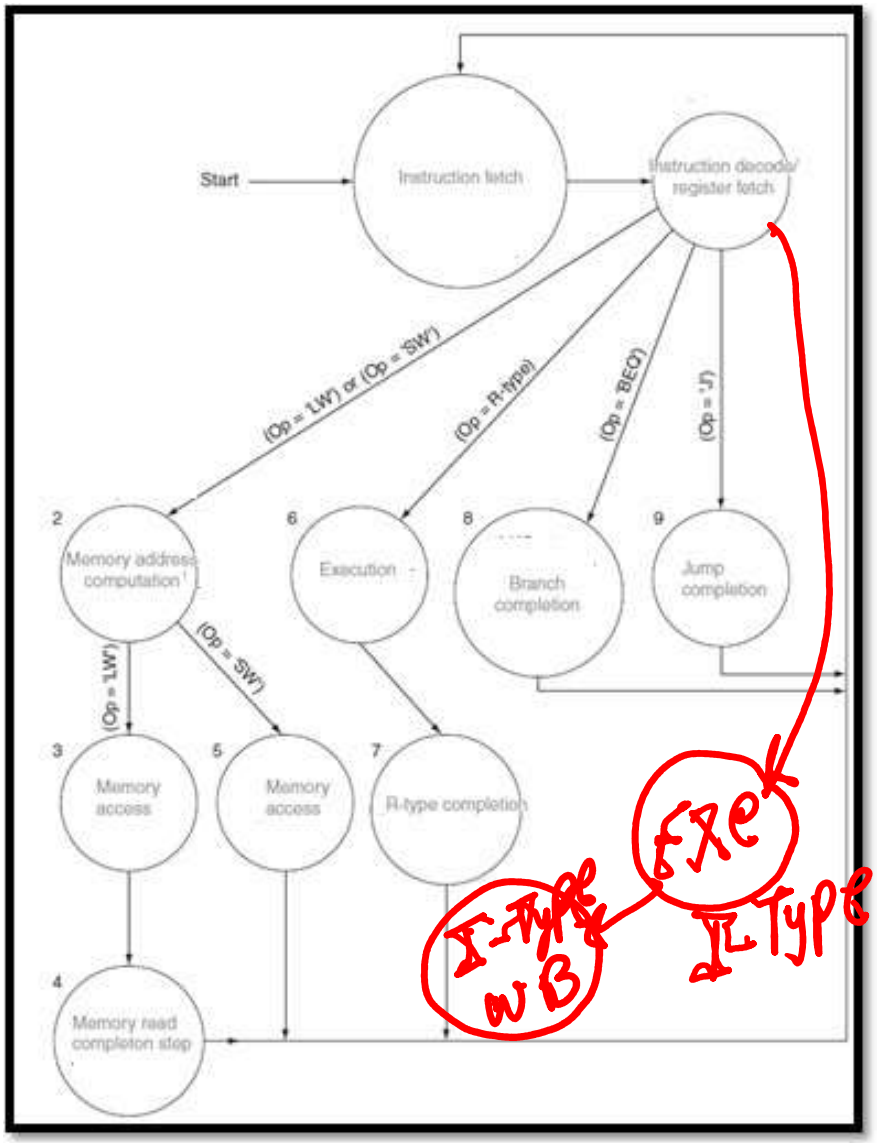
MIPS Multicycle CPU FSM Controller

Datapath and Control

MIPS Multicycle CPU FSM



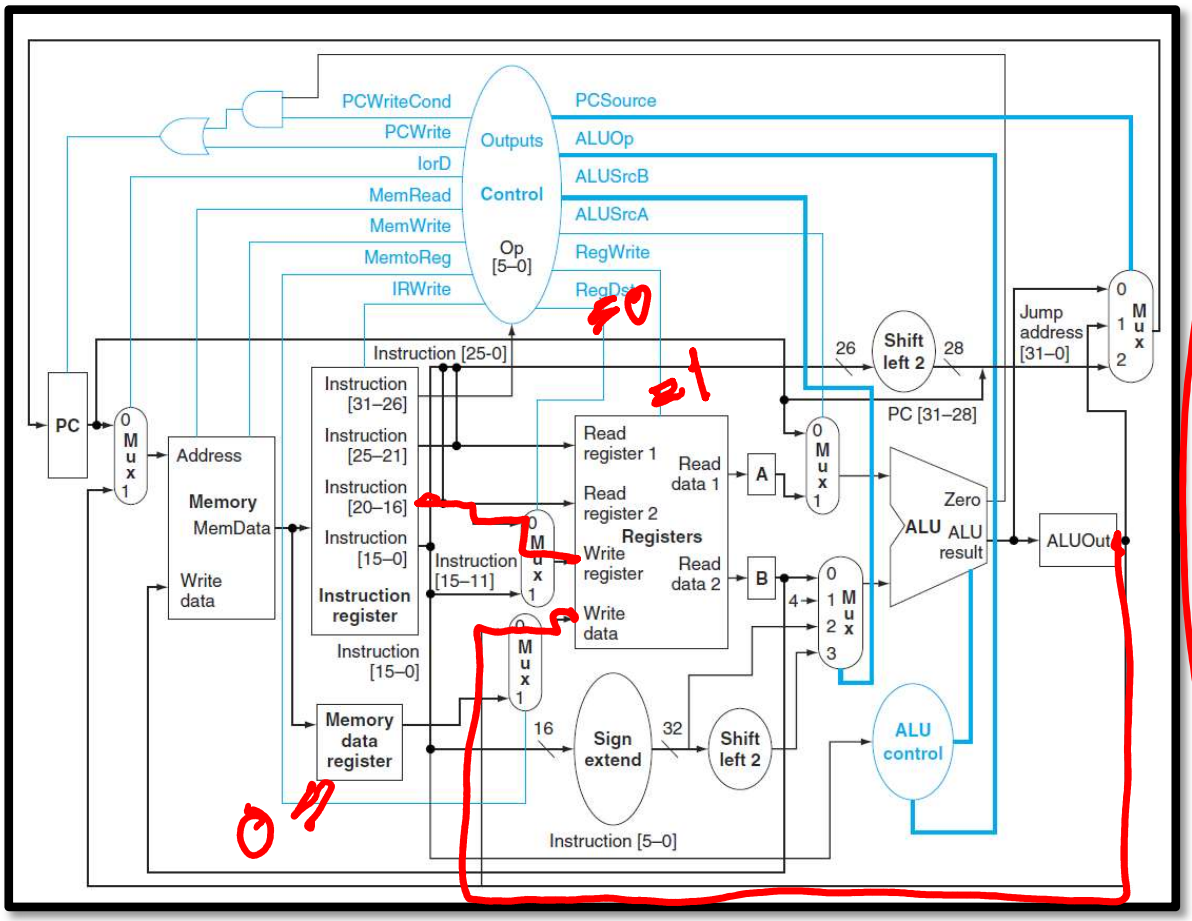
MIPS Multicycle CPU Datapath



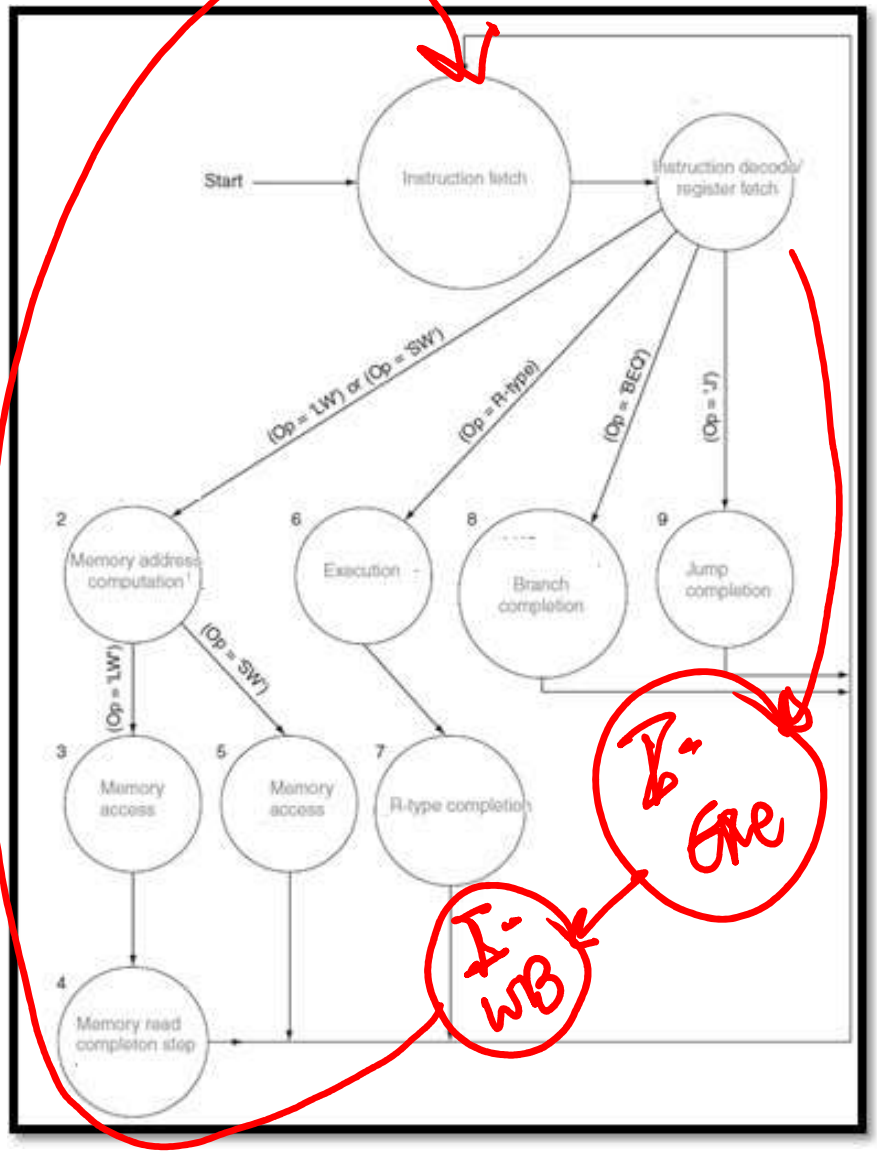
MIPS Multicycle CPU FSM Controller

Datapath and Control

MIPS Multicycle CPU FSM



MIPS Multicycle CPU Datapath



MIPS Multicycle CPU FSM Controller

1. give rst
2. assign your clock
3. have your instr ready @ memdatain

3- Force r_2 = Some value
make sure that WB register is not r_2

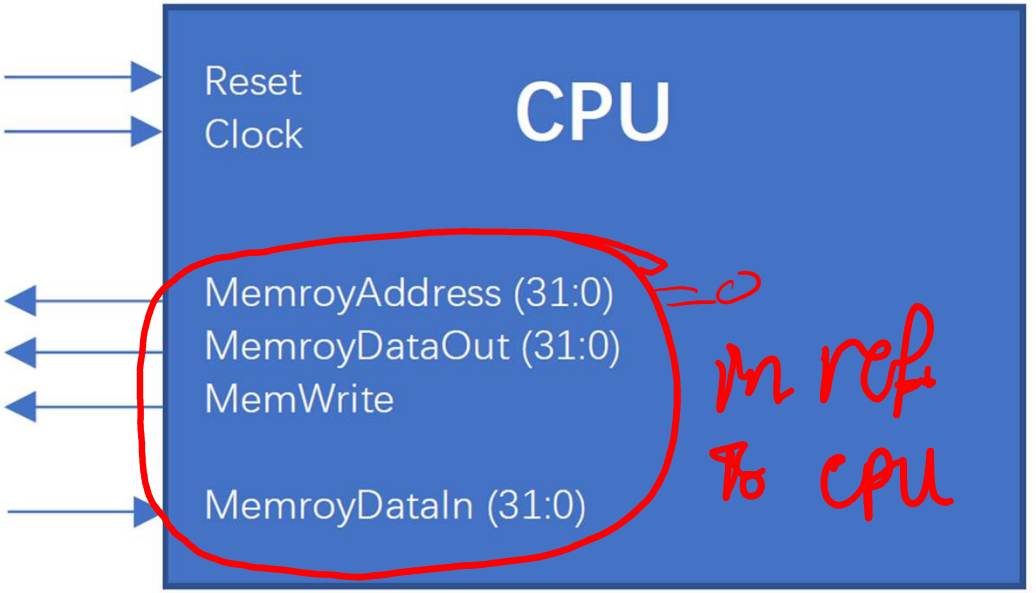


Figure 2: CPU top level structure

4. rst = 0
check MemAddress = 0
MemWrite = 0

ADDI $r_3, r_2, 4$

5. wait 1 clockcycle to pass
chk MemAddress = 4

6. Wait for 3 CC to pass
check result in your reg

$r_3 = 7$